

Hospital Emergency Room Data Quality & Preparation Report

Project Objective

The objective of this project is to analyze Emergency Room (ER) operations and identify opportunities to improve patient experience, operational efficiency, and quality of care through data-driven insights.

Business Questions

What drives long patient wait times?

What drives low patient satisfaction?

What factors contribute to patient readmissions?

Which departments experience the highest operational pressure?

How can hospital management improve Emergency Room performance?

Dataset Overview

Dataset Name

Hospital Emergency Room Operations Dataset

Original Records

10,130 Emergency Room Visits

Final Cleaned Records

10,100 Emergency Room Visits

Fact Table

Fact_ER_Visits

Dimension Tables

Dim_Department

Dim_Physician

Dim_Diagnosis

Dim_ReferralSource

Initial Data Quality Assessment

The raw dataset was audited before analysis to identify data quality issues that could affect reporting accuracy and business insights.

The following issues were identified:

Missing Values

Department

Referral Source

Chief Complaint

Attending Physician

Satisfaction Score

Gender
Triage Level
Readmission
Inconsistent Categories
Admission Flag
Readmission Status
Triage Level
Shift
Gender
Race
Physician Names
Invalid Values

Wait Time contained impossible operational values.

Minimum Wait Time: -5 minutes
Maximum Wait Time: 999 minutes

These values were identified as data-entry errors and required correction before analysis.

Data Quality Findings

Missing Values

Field Missing Records

Department 5,523

Referral Source 1,808

Chief Complaint 997

Attending Physician 859

Satisfaction Score 7,088

Inconsistent Admission Flag Values

The Admission Flag contained multiple representations of the same business concept, including:

Yes

No

TRUE

FALSE

Admitted

Admission

This inconsistency created duplicate categories during reporting.

Inconsistent Readmission Values

The Readmission field contained:

Yes

No

TRUE

FALSE

N

0

Inconsistent Shift Values

Examples included:

Morning

morning

MORNING

Morning Shift

Afternoon

AFTERNOON

Night

Night Shift

Inconsistent Gender Values

Examples included:

M

F

Male

Female

Maleale

Femaleemale

Femaleemaleemale

Inconsistent Race Values

Race categories contained inconsistent capitalization, spacing, and formatting.

Physician Names

Physician names contained inconsistent capitalization and formatting.

Invalid Wait Time Values

A quality audit identified:

16 records with negative wait times

14 records with a placeholder value of 999 minutes

These 30 records represented 0.30% of the dataset and were classified as invalid operational values.

Data Cleaning Log

Step 1

Standardized Admission Flag

Before

Yes

yes

TRUE

FALSE

Admitted

Admission

No

no

After

Yes

No

Reason

Standardized admission values to eliminate duplicate reporting categories and ensure consistent operational analysis.

Step 2

Standardized Readmission Status

Before

Yes

TRUE

No

FALSE

N

0

After

Yes

No

Reason

Prevent duplicate categories and improve reporting accuracy.

Step 3

Standardized Triage Levels

Before

1

2

3

4

5
L1
L2
L3
L4
L5
ESI 1
ESI 2
ESI 3
ESI 4
ESI 5
Level 1
Level 2
Level 3
Level 4
Level 5
After
Level 1
Level 2
Level 3
Level 4
Level 5
Reason

Different naming conventions represented the same triage severity levels and created duplicate reporting categories.

Step 4

Standardized Shift Values

Before

Morning

morning

MORNING

Morning Shift

Afternoon

AFTERNOON

Afternoon Shift

Night

Night Shift

After

Morning

Afternoon

Night

Reason

Ensure consistent operational reporting across work shifts.

Step 5

Standardized Gender Values

Before

M

F

Male

Female

Maleale

Femaleemale

Femaleemaleemale

After

Male

Female

Reason

Correct data entry inconsistencies and eliminate duplicate categories.

Step 6

Standardized Race Values

Actions Performed

Trim

Clean

Capitalize Each Word

Example

Before

AFRICAN AMERICAN

african american

African American

After

African American

Reason

Remove duplicate race categories caused by inconsistent capitalization and spacing.

Step 7

Standardized Physician Names

Actions Performed

Trim

Clean

Capitalize Each Word

Example

Before

DR. MARIA JACKSON

dr. maria jackson

Dr. Maria Jackson

After

Dr. Maria Jackson

Reason

Ensure accurate physician-level reporting.

Step 8

Missing Value Treatment

Field - Action Taken

Department - Replaced with "Unknown"

Referral Source - Replaced with "Unknown"

Chief Complaint - Replaced with "Other/Unspecified"

Attending Physician - Replaced with "Unknown"

Gender Replaced with "Unknown"

Triage Level Replaced with "Unknown"

Readmission Replaced with "Unknown"

Satisfaction Score Retained as Blank

Reason

Missing satisfaction scores represent the absence of patient feedback rather than poor satisfaction. Therefore, blank values were preserved and excluded from satisfaction calculations.

Step 9

Wait Time Quality Review

The wait-time field was audited to identify operationally impossible values.

Records identified:

16 negative wait times

14 placeholder values of 999 minutes

Total invalid records:

30 records (0.30% of the dataset)

These records were excluded from wait-time analysis to improve analytical accuracy while preserving the integrity of the remaining dataset.

Step 10

Date Standardization & Feature Engineering

Columns Reviewed

Visit Date

Discharge Date

Issue

The dataset contained mixed date formats, including Excel serial dates and DateTime values.

Actions Performed

Standardized Visit Date

Standardized Discharge Date

Created Visit Year

Created Visit Month

Created Visit Day of Week

Created Visit Hour

Reason

Enable time based trend analysis and operational reporting.

Data Cleaning Summary

Metric Result

Original Records 10,130

Final Cleaned Records 10,100

Invalid Wait Time Records Removed 30

Missing Departments Replaced 5,523

Missing Referral Sources Replaced 1,808

Missing Chief Complaints Replaced 997

Missing Physicians Replaced 859

Missing Satisfaction Scores Retained 7,088

Admission Categories Standardized Yes

Readmission Categories Standardized Yes

Triage Levels Standardized Yes

Shift Categories Standardized Yes

Gender Categories Standardized Yes

Race Categories Standardized Yes

Physician Names Standardized Yes

Date Features Created Year, Month, Day of Week, Hour

Business Impact of Data Cleaning

The data preparation process significantly improved the quality and reliability of the dataset for business reporting.

Key benefits include:

Eliminated duplicate reporting categories through standardized categorical values.
Improved operational reporting accuracy by removing invalid wait-time records.
Preserved valuable records by replacing missing categorical values with standardized placeholders.
Maintained analytical integrity by retaining missing satisfaction scores as blank rather than introducing artificial values.
Enabled monthly, daily, hourly, and seasonal trend analysis through feature engineering.
Produced a clean and consistent dataset suitable for KPI reporting, dashboard development, and executive decision-making.

Conclusion

A comprehensive data quality assessment was completed before analysis. Data inconsistencies, missing values, and invalid records were identified, documented, and addressed using appropriate cleaning techniques while preserving business meaning.

The resulting cleaned dataset contains 10,100 Emergency Room visits and provides a reliable foundation for analyzing patient wait times, satisfaction, admissions, demographics, physician performance, departmental operations, and overall Emergency Room efficiency.